

Mental Health Metrics

Executive Summary

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Background

Mental Health Metrics (MHM) is a Pittsburgh-based organization focused on early detection of schizophrenia relapse using a validated prediction algorithm. The organization is working to develop technology-based tools that address the limitations of traditional post-hospitalization treatments to significantly reduce the number of relapses, and their associated costs. Core analytical work is performed with an Excel-based workflow, including client survey analysis and report generation. This approach has been sufficient to validate MHM's proprietary relapse prediction algorithm, and demonstrate proof of concept. While this strategy has strengthened the foundation of the research, it has delayed development of technological infrastructure necessary to attract research funding and support future demonstrations.

Project Description

Project Opportunity

MHM's primary organizational challenge is the gap between its validated algorithmic capabilities and its current technology infrastructure. Relapse risk analysis and report generation rely on a manual, Excel-based workflow. While the Excel workflow was sufficient to demonstrate proof of concept through pilot studies, the process leaves MHM without the software infrastructure needed to carry out a funded study effectively. As MHM's mission depends on the timely identification of relapse risk, delays or inconsistencies in data processing reduces clinicians' ability to intervene, undermining the true preventative value of the technology. MHM has the opportunity to develop its relapse prediction algorithm into a prototype that supports research studies to attract funding and test the tool on a larger scale.

Project Vision

A minimum viable product (MVP) was developed that automates MHM's relapse detection workflow from survey data ingestion to risk reporting. This solution prioritizes accuracy and scalability, helping MHM move from a research-proven algorithm to a deployable tool. Ultimately, the MVP serves as a critical step toward securing funding for a demonstration project with 100 clients across multiple sites, gathering essential data before the product can be marketed and adopted more broadly by mental health care providers.

Project Outcomes

A data pipeline was built that connects Google Sheets, Google App Scripts, and analysis code, enabling automated data flow between system components for downstream processing. Through the MVP, the client's relapse risk algorithm was defined more concisely, through statistical "modules" to prevent redundant code. This was reflected using MVP components that aligned with client approved use cases and decisions, resulting in automated data analysis without manual intervention. Flow diagrams and other forms of documentation were created to strengthen client understanding. Documents were provided containing information on the MVP's structure, data flow, and report interpretation for future development.

Project Deliverables

Mental Health Metrics will be provided with a complete implementation of a minimum viable product that can be used for testing. This includes a Google account with Google App Scripts, Forms, and Sheets to collect and process data for the algorithm. Data analysis is done with Python in a GitHub repository. The entire minimum viable product's workflow, structure, code, and report interpretations are documented for future understanding and reference.

Recommendations

A robust cloud storage system is recommended, improving reliability and scalability when maintaining larger amounts of patient data. This would maintain data integrity and support broader clinical adoption. Further down the line, developing a web-based application is recommended, incorporating the risk algorithm into an accessible interface for patients and clinicians. Patients could complete forms directly through the platform, and clinicians could log in to view reports and track risk levels in one centralized location. This portal could also integrate an AI "pal" to communicate and interact with patients, assisting with adherence with the program and ensuring treatment is as successful and accurate as possible.

Student Consulting Team

Megha Jasti served as the client liaison. She is a third-year student majoring in Information Systems with an additional major in Business Administration, graduating in Fall 2026. She will be interning at PricewaterhouseCoopers this summer as a Corporate Tech Strategy intern, and is looking toward a career in technology strategy consulting.

Jasmine Shi was the product and technical lead. She is a third-year student majoring in Information Systems with an additional major in Human-Computer Interaction, graduating in December 2026. She will be at Yes Energy this summer as an AI Product Management Intern and is pursuing a career in product management and technology strategy.

Sofia Yu led the software development. She is a third-year student majoring in Information Systems, with a minor in Computer Science and a concentration in AI Management, graduating in Spring 2027. She will be joining Atlassian as a Software Engineering Intern in the upcoming summer, pursuing a career in software engineering.